Université Gustave Eiffel
M2 – Semester 2 – 2025/2026

Advanced Data Science - Homework 3

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Exercise 1. Let $\mathcal{G}$ be a class of classifiers and

$$\mathcal{H}_\lambda = \left\{ \sum_{j=1}^{m} \alpha_j g_j : m \in \mathbb{N}, \boldsymbol{\alpha} \in \mathbb{R}_+^m, (g_1, \dots, g_m) \in \mathcal{G}^m, \sum_{j=1}^{m} \alpha_j \leq \lambda \right\}.$$

Show that $\mathcal{H}_\lambda$ is convex.

Solution. Let $h, k \in \mathcal{H}_\lambda$. Then there exist $m, n \in \mathbb{N}$ and $\boldsymbol{\alpha} \in \mathbb{R}_+^m$, $\boldsymbol{\beta} \in \mathbb{R}_+^n$, $(g_1, \dots, g_m) \in \mathcal{G}^m$, $(g'_1, \dots, g'_n) \in \mathcal{G}^n$ such that

$$h = \sum_{j=1}^{m} \alpha_j g_j, \sum_{j=1}^{m} \alpha_j \leq \lambda, \quad k = \sum_{j=1}^{n} \beta_j g'_j, \sum_{j=1}^{n} \beta_j \leq \lambda.$$

Let $\theta \in [0, 1]$. Then

$$\theta h + (1-\theta)k = \sum_{j=1}^{m} \theta \alpha_j g_j + \sum_{j=1}^{n} (1-\theta)\beta_j g'_j.$$

We have $(g_1, \dots, g_m, g'_1, \dots, g'_n) \in \mathcal{G}^{m+n}$ and $(\theta\boldsymbol{\alpha}, (1-\theta)\boldsymbol{\beta}) \in \mathbb{R}_+^{m+n}$, we have $\theta h + (1-\theta)k \in \mathcal{H}_\lambda$. Moreover,

$$\sum_{j=1}^{m} \theta \alpha_j + \sum_{j=1}^{n} (1-\theta)\beta_j = \theta \sum_{j=1}^{m} \alpha_j + (1-\theta) \sum_{j=1}^{n} \beta_j \leq \theta\lambda + (1-\theta)\lambda = \lambda.$$

Thus $\mathcal{H}_\lambda$ is convex.